

Agentic AI Control Plane:

From Frontier Models to Reliable Enterprise Workflows

Evals · Guardrails · Tool Governance · Observability · Human Oversight · Auditability

Raj Sundaramoorthy · Applied AI Engineering Leader

Multi-Agent Systems · RAG · Evals · Guardrails · Observability · github.com/rbsundaramoorthy

Built for regulated and high-trust workflows across the enterprise.

CORE THESIS

Agents are not production-ready because they can call tools. They become production-ready when every action is governed by intent, identity, retrieval boundaries, evaluation, traceability, human oversight, and audit evidence.

REFERENCE CONTROL PLANE ARCHITECTURE

01 Intent & Workflow

Trigger policy · use-case gating · state machine · escalation path

02 Context & Retrieval

ACL-safe RAG · source ranking · freshness · citation boundaries

03 Tool Governance

Allowed actions · approval gates · safe-action limits · secrets control

04 Agent Runtime

Planner · tool router · memory · structured outputs · fallback strategy

05 Evals & Guardrails

Ground-truth tests · confidence scoring · regression gates · policy checks

06 Observability

LLM/tool traces · latency · cost · quality · drift · replayable evidence

CONTROL PLANE OVERLAY

Identity · policy · risk tiering · audit trail · human review · deployment gates

WHAT THE CONTROL PLANE GOVERNS

- Use-case intake: classify workflow risk, persona, data sensitivity, and tool blast radius.
- Governed retrieval: tenant and ACL-safe access; no answer without grounding when policy requires.
- Safe tool use: allowlist tools, constrain parameters, require human approval for high-impact actions.
- Evaluation layer: golden datasets, fuzzy matching, regression tests, confidence thresholds, hallucination checks.
- Runtime observability: trace prompts, model and tool calls, retrieved context, cost, latency, errors, decisions.
- Audit evidence: replayable run history of what the agent knew, decided, used, and produced.

WHY I AM CREDIBLE FOR THIS WORK

- Accenture: lead a 7-person AI engineering team shipping agentic dev tooling, MCP workflows, RAG, evals, guardrails, and Responsible AI controls.
- Outcomes: 22% faster code-review turnaround; utility agents recover about an hour per employee per day.
- TIAA: built observability and data-platform architecture for retirement-income workflows with daily ledgering, reconciliation, and regulatory reporting.
- The Hartford: founded two teams from zero; built Entity Resolution as a Service; shipped ML claims analytics saving \$6.2M per year.
- GitHub proof: a multi-agent due-diligence analyst with claim-level provenance, per-claim confidence, source tiering, evals, and full run traces, plus an enterprise RAG platform and a streaming journey-intelligence agent.

WHERE THIS APPLIES

Enterprise copilots

Financial-services agents

Advisor / customer workflows

Research and due-diligence agents

Internal productivity agents

Regulated decision-support workflows

BEST-FIT TEAMS

Applied AI · Agentic AI Platform · Enterprise AI · AI Governance · AI Solutions · FS AI Transformation

OPERATING MODEL

Ship one high-value workflow first. Prove it with evals, observability, human review, and adoption. Then scale the patterns into platform capabilities.

SUCCESS METRICS

Task quality · human acceptance rate · answer grounding · tool success and failure · risk exceptions · latency · cost · adoption · business KPI movement.

WHY THIS MATTERS TO AI LABS

Applied AI teams need repeatable patterns that help customers move from impressive demos to reliable, observable, governed production workflows.